

Who Leaves a Recorded Path into Government? Structure, Documentation, and Distribution Shift in CBDB

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ABSTRACT

Prosopographical databases make collective historical analysis possible, but their records reflect source survival, editorial priorities, and database encoding. We predict whether each of 661,124 China Biographical Database (CBDB) people has an ENTRY_DATA record. This operational label, E , differs from both posting-record presence, P , and the latent state of true historical entry, T . We therefore estimate retrospective record presence rather than historical truth. Frozen logistic and CatBoost models were evaluated through grouped ablation, paired bootstrap, five-seed stability, family and spatial shifts, validation-fitted calibration, and grouped SHAP. The designated D5_MAIN model reached Global ROC-AUC 0.936956 and PR-AUC 0.882535. Conditional evidence was strongest for gender, address observability, administrative geography, and family observability, whereas recorded family political capital added little after observability and topology. Discrimination weakened on matched-support unseen historical regions. Full-coverage source-group confirmation was infeasible under the prespecified connected-component protocol because 448,632 of 452,035 eligible people belonged to a single connected component. These results show that CBDB ENTRY-record presence is highly predictable, while also locating that performance within the database's documentation and selection processes. They do not reconstruct true historical entry or identify causal effects.

CCS CONCEPTS

• **Computing methodologies** → **Supervised learning by classification**; • **Applied computing** → *Digital libraries and archives*.

KEYWORDS

computational history, prosopography, documentation bias, distribution shift, CBDB

1 INTRODUCTION

Prosopographical databases convert dispersed biographical traces into relational data for collective, spatial, and network analysis [1, 2, 4]. CBDB is a prominent example, combining structured biographies with relations, places, offices, kinship, and texts [5, 10, 20]. Such databases are not random censuses of past populations. Their contents depend on which people entered surviving sources, which sources were processed, and how ambiguous statements were encoded.

This study combines open-ended exploratory analysis of dynastic, gender, ENTRY-pathway, geographic, and kin-observability patterns with predictive modeling of ENTRY-record presence.

This selection creates a specific prediction problem. ENTRY-record presence may reflect historical structure, including period, geography, and family relations. It may also reflect the documentation process that made those attributes and the label observable. Consequently, a high area under the receiver-operating-characteristic curve (ROC-AUC) is insufficient evidence of historical recovery. The central problem is to separate predictive structure from record-generation signals and to state where that separation fails.

We ask five questions. **RQ1** asks whether ENTRY-record presence can be predicted reliably on a frozen person-level test. **RQ2** asks how personal, geographic, family, and documentation blocks contribute conditionally. **RQ3** asks whether recorded family political capital adds information after observability and topology. **RQ4** asks how the frozen F2 structural comparator changes under whole-family grouping and how locked models transport to unseen historical regions. **RQ5** asks how far high performance reflects historical structure versus database recording processes.

The study contributes a reproducible measurement and leakage policy, a frozen three-level model hierarchy, and an evidence-ordered interpretation. H_STRUCT is a historical structural benchmark, although it still contains address and kin observability. D5_MAIN is the designated predictive model without relatives' outcomes. D6_UPPER is a database-internal record-structure upper bound. Grouped ablation and distribution shifts carry the main interpretation; SHAP is retained only as subordinate attribution evidence.

2 RELATED WORK

Computational prosopography. CBDB supports statistical, network, spatial, kinship, and reference analysis of heterogeneous historical texts [1, 2, 4, 20]. Its official guide documents schema and export procedures, not complete historical coverage [5].

Prediction and generalization. Tree ensembles are strong heterogeneous-tabular baselines, and CatBoost limits target leakage in categorical statistics [7, 16]. Random holdout can be optimistic under structured dependence, while covariate shift can invalidate ordinary validation [17, 19]. WILDS and transport research motivate explicitly bounded shift tests [13, 18].

Measurement, attribution, and calibration. Administrative data inherit selection processes; operational measures need not equal latent constructs, and imperfect labels alter classification evidence [3, 9, 12]. Dataset documentation should expose these boundaries [6]. SHAP is neither context-free importance nor causal evidence, and calibration requires held-out fitting data [8, 11, 14, 15].

3 DATA AND MEASUREMENT PROBLEM

The frozen input is official release `cbdb_20260829.sqlite3`. The assignment brief reports approximately 515,488 people, whereas the verified release used in this study contains 661,124 BIOG_MAIN person records. CBDB counts are release-specific; all results here refer only to `cbdb_20260829.sqlite3`. CBDB is a selectively assembled historical database rather than a population register. Relation tables were aggregated by person before joining to BIOG_MAIN, which prevented multi-table fan-out. The resulting analysis population contains 661,124 covered people and 220,627 positive ENTRY labels (Table 1).

ENTRY_DATA includes examination or degree records, school or student status, recommendation, hereditary or *yin* privilege, military entry, purchase or donation, direct appointment, and other or ambiguous entry records. Define $E_i = 1$ when person i has at least one ENTRY row and $P_i = 1$ when the person has at least one valid posting row. Let T denote the latent true historical state of entry into government. At the construct level,

$$E \neq T, \quad P \neq T, \quad E \neq P. \quad (1)$$

E , P , and T are distinct constructs and are not semantically equivalent. Thus $E = 0$ means that no ENTRY row was found, while $E = 1$ does not necessarily imply actual office holding. Posting is also incomplete and cannot serve as historical ground truth. The estimand is $\Pr(E_i = 1 \mid X_i)$.

Figure 1b makes the proxy boundary concrete. The $E = 1, P = 1$ quadrant records both an ENTRY and a posting trace. The $E = 1, P = 0$ quadrant records ENTRY without a valid posting, while $E = 0, P = 1$ records posting without ENTRY. The $E = 0, P = 0$ quadrant records neither relation. None of these quadrants identifies T without additional historical evidence.

4 TASK DEFINITION AND LEAKAGE CONTROL

The primary task classifies retrospective ENTRY-record presence. Direct ENTRY fields, the focal person’s posting outcomes, entry or posting counts, raw index year, and person or family identifiers are forbidden predictors. Safe birth information requires audited provenance. Regional density is fitted on training people only. The local ENTRY prior groups the exact `addr_id` key, applies five-fold deterministic out-of-fold encoding with smoothing strength 20 on training people, and transforms validation or test people from the complete-training mapping; unseen keys use the relevant training mean. Test labels select neither features nor hyperparameters, thresholds, early stopping, or calibration.

The model hierarchy isolates progressively more database structure (Table 2). `H_STRUCT` contains regime, gender, safe birth cohort, geography, address availability, and kin observability or topology. It is therefore not an unbiased historical model. `D5_MAIN` adds documentation-linked features and a supervised local prior but excludes relatives’ outcomes. `D6_UPPER` adds full-record relatives’ lifetime ENTRY and posting summaries. Those temporally ambiguous fields make `D6_UPPER` an upper bound rather than a historical mechanism model.

5 EXPLORATORY DATA ANALYSIS

Across Tang, Song, Yuan, Ming, and Qing, CBDB coverage ranged from 25,310 to 237,423 people and E prevalence from 3.17% to 54.54% (Figure 2). Historical institutions, corpus composition, database projects, and source survival remain inseparable explanations.

Recorded gender is also imbalanced: 87.55% of covered people are male, and E prevalence is 37.96% for men, 1.22% for women, and 0.79% for unknown gender. This predictive partition is not a historical comparison of opportunity.

ENTRY pathways vary across dynasties (Figure 3). Examination, schooling, recommendation, privilege, military, and appointment categories are nonexclusive at person level. Their heterogeneity may reflect historical access, terminology, or source processing.

Observability differs by feature: addresses cover 59.83% of people, any kin 43.29%, core family 33.94%, valid coordinates 57.82%, and safe birth years 9.05%. Documentation intensity also varies by population, allowing models to learn record production alongside historical attributes.

6 FEATURE CONSTRUCTION

Features separate personal attributes; address observability, administrative and physical geography, density, semantics, and local prior; family observability, topology, and capital; and general documentation flags.

Administrative identifiers mix place and normalization; address and kin flags directly measure observability; family topology is partly transductive; and relatives’ outcomes are not strictly pre-entry. Grouped ablations preserve these boundaries.

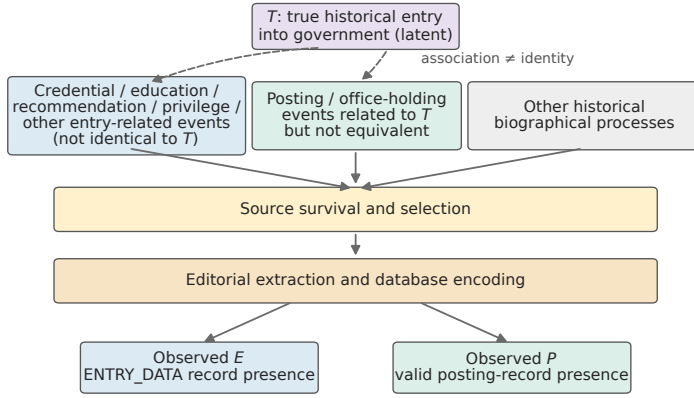
7 MODELS AND EVALUATION PROTOCOL

Balanced logistic regression is the frozen linear baseline. CatBoost models use seed 42, at most 500 iterations, depth 7, learning rate 0.05, L_2 leaf regularization 5, and validation-only early stopping. No Phase 3.1 model was trained or tuned. The primary split is a frozen, person-level, dynasty-by- E stratification. Whole-family robustness keeps connected family groups disjoint. Matched-support spatial tests compare random and unseen-region tests within the same reliable-prefecture support, although their test people differ.

We report ROC-AUC, precision-recall AUC (PR-AUC), balanced accuracy, F1, log loss, Brier score, and expected calibration error (ECE). PR-AUC is interpreted with test prevalence. The sigmoid calibrator was fitted on validation predictions only. Table 3 labels raw log loss and Brier score separately from validation-calibrated ECE. Paired bootstrap used 500 stratified resamples only when the exact prediction pair was retained. These intervals quantify test-sample variability, not uncertainty in label validity, source survival, or inclusion in CBDB.

Five prespecified seeds assess stability for selected contrasts. The SAFE temporal split is reported as a split diagnostic because no frozen locked-model performance artifact exists. Qing is descriptive because no prespecified Qing holdout was frozen. A source-group model was conditional on a feasibility gate and was not fitted because full-coverage confirmation was infeasible under the prespecified full-coverage connected-component protocol.

a Constructs, historical events, and recorded labels



E, P, and T are distinct constructs and are not interchangeable.

b Frozen $E \times P$ contingency

Observed ENTRY label E	0	1
1	85,255	135,372
0	276,858	163,639
Observed Posting label P		
		0 1

Figure 1: Constructs, historical events, and recorded labels. a, T is latent true historical entry. Credential, education, recommendation, privilege, and other entry-related events are not T ; posting or office-holding events are related to, but not equivalent to, T . Source survival, extraction, and encoding produce the observed E and P relations. E , P , and T are distinct constructs and are not interchangeable. b, Frozen contingency for all 661,124 CBDB-covered people.

Table 1: Dataset and observed targets. Counts and shares are generated from the frozen person target and E-P contingency tables.

Quantity	Value	Operational definition
CBDB-covered people	661,124	One row per BIOG_MAIN person
E positives	220,627	At least one ENTRY_DATA row
E prevalence	0.3337	Recorded ENTRY share among CBDB-covered people
P positives	299,011	At least one valid posting record
Primary-test people	99,307	Frozen dynasty-by-E stratified test

Table 2: Locked model roles and leakage boundary. Feature counts are read from the Phase 2.6 lock artifact.

Model	Features	Role	Boundary
H_STRUCT	26	Historical structural benchmark	No general documentation, address type, local target prior, or relative outcomes
D5_MAIN	34	Main predictive model	Documentation and topology; no relative outcomes
D6_UPPER	44	Record-structure upper bound	D5 plus full-record relative ENTRY/posting outcomes

8 EXPERIMENTAL RESULTS

8.1 Locked models establish strong within-database discrimination

Table 3 and Figure 4 establish the frozen hierarchy. Global D5_MAIN achieved ROC-AUC 0.936956 and PR-AUC 0.882535, with raw log loss 0.315106 and raw Brier score 0.097356. Its validation-calibrated ECE was 0.003438. Performance also remained high in Song and Ming, although population prevalence and PR-AUC differed.

D6_UPPER improved only marginally over D5_MAIN. Because it includes relatives' lifetime ENTRY and posting information, the

comparison bounds record-structure predictability rather than supporting a stronger historical interpretation. H_STRUCT performed below D5_MAIN but remained strong, showing that regime, geography, family structure, and their observability already contain substantial predictive signal.

8.2 Ablation separates observability from family capital

Grouped ablation is the primary evidence of conditional information (Figure 5; Table 4). Global increments were largest for gender,

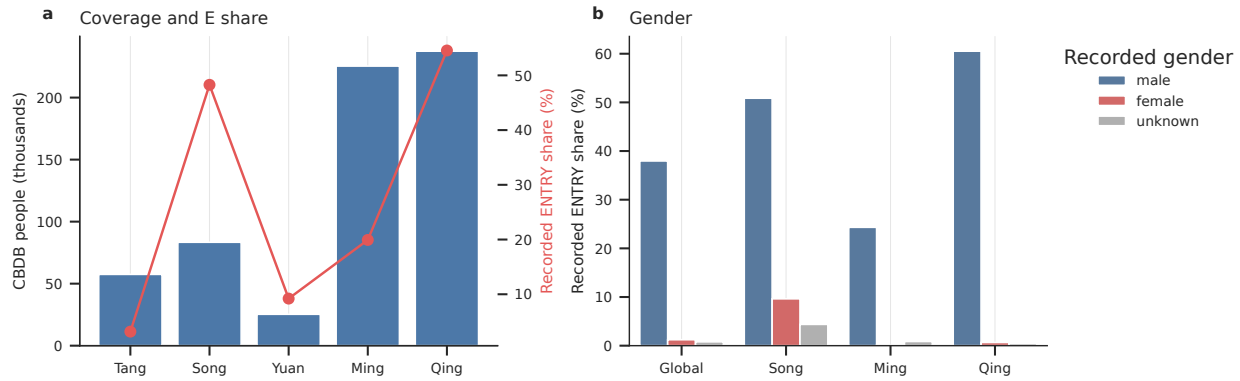


Figure 2: CBDB coverage and recorded ENTRY share. a, Covered people and E prevalence within Tang, Song, Yuan, Ming, and Qing. b, E prevalence within recorded gender groups for Global, Song, Ming, and Qing. Denominators are CBDB-covered people, not historical populations.

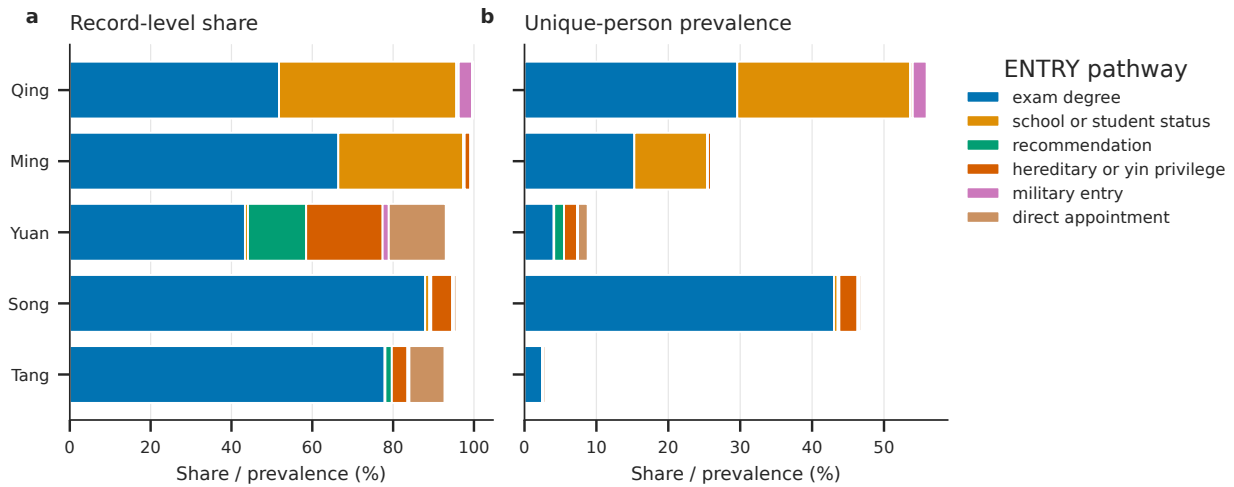


Figure 3: Recorded ENTRY pathway composition. a, Share of ENTRY records by semantic category and dynasty. b, Prevalence of each category among all CBDB-covered people in that dynasty. Person-level pathways are nonexclusive and may sum above 100%.

address observability, administrative geography, and family observability. Address-record semantics added a smaller but visible increment. Physical geography changed ROC-AUC by 0.000014 and PR-AUC by -0.000343. Its exact A2/A3 paired predictions were not retained, so no valid confidence interval is reported. Because the grouped ablations follow a prespecified nested order, every increment is conditional on the blocks already entered. The increments are therefore order-dependent and cannot be interpreted as unique contributions or causal effects.

Family observability increased Global ROC-AUC by 0.0217, while topology added 0.0023 (95% paired bootstrap interval 0.0019 to 0.0026). Full-record family capital added 0.0007 (0.0005 to 0.0009), and train-observed capital added 0.0003 (0.0001 to 0.0005). These positive increments are stable but substantively small after observability and topology. Five-seed ranges in Appendix Figure A2 support the same ordering.

8.3 Family and spatial shifts bound generalization

Address decomposition separates a large observability increment from administrative geography, address semantics, and near-zero physical or local-prior increments (Figure 6a). This ordering shows conditional predictive information, not geographic causation. The matched-support test then provides a distinct boundary: unseen-region ROC-AUC decreased for H_STRUCT and D5_MAIN in Global, Song, and Ming (Figure 6b). Global decreases were 0.0307 and 0.0222, respectively.

Whole-family robustness was evaluated only for the frozen F2 structural comparator in Global and Ming. Family-group robustness of D5_MAIN was not directly evaluated. Globally and in Song, family-capital increments were small and mostly positive. The full-record F3-F2 ROC-AUC increment was +0.000701 globally and +0.001821 in Song. In Ming, the full-record increment was -0.000319

Table 3: Frozen primary-test performance. Log loss and Brier score use raw model probabilities; ECE uses probabilities from a sigmoid calibrator fitted on validation predictions only. The logistic artifact did not retain a directly comparable calibrated ECE. D5_MAIN is the designated main result.

Population	Model	ROC-AUC	PR-AUC	Raw LogLoss	Raw Brier	Validation-calibrated ECE
Global	Logistic M6	0.9160	0.8436	0.3635	0.1128	–
Global	H_STRUCT	0.9267	0.8582	0.3393	0.1048	0.0036
Global	D5_MAIN	0.9370	0.8825	0.3151	0.0974	0.0034
Global	D6_UPPER	0.9374	0.8831	0.3140	0.0969	0.0034
Song	Logistic M6	0.9008	0.8810	0.3985	0.1223	–
Song	H_STRUCT	0.9183	0.9116	0.3604	0.1112	0.0072
Song	D5_MAIN	0.9384	0.9398	0.3089	0.0940	0.0094
Song	D6_UPPER	0.9398	0.9412	0.3055	0.0928	0.0102
Ming	Logistic M6	0.9021	0.7765	0.3500	0.1126	–
Ming	H_STRUCT	0.9107	0.7799	0.3463	0.1097	0.0029
Ming	D5_MAIN	0.9164	0.8005	0.3229	0.1026	0.0027
Ming	D6_UPPER	0.9167	0.8012	0.3224	0.1023	0.0025

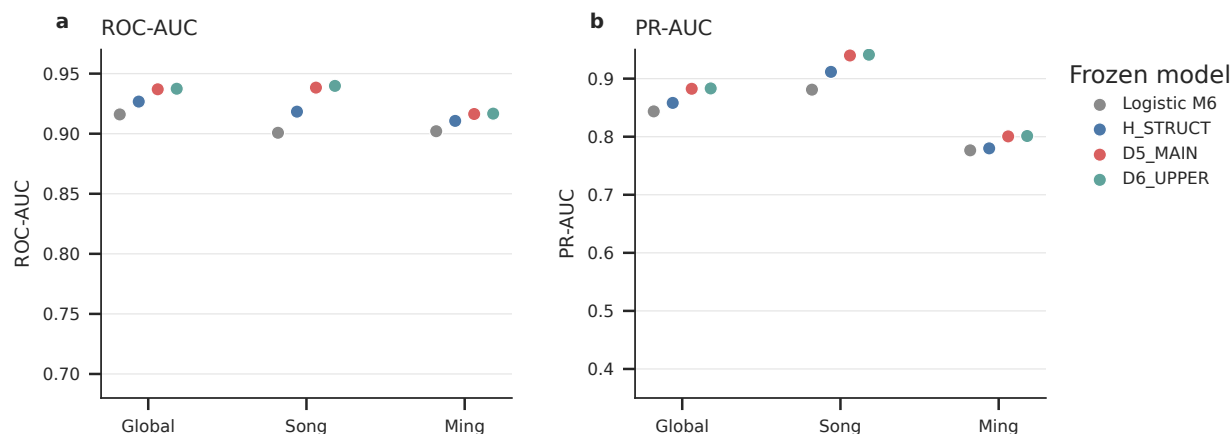


Figure 4: Frozen primary-test discrimination. ROC-AUC and PR-AUC compare the logistic baseline, H_STRUCT, D5_MAIN, and D6_UPPER in Global, Song, and Ming. D5_MAIN is the designated main model; D6_UPPER is a database-internal upper bound.

Table 4: Global grouped ablation evidence. Point estimates are frozen conditional increments. Intervals are 500-resample paired bootstrap intervals only where the exact paired prediction pair was retained. Dashes, including Physical Geography, mean that no valid exact-pair interval is available, not zero uncertainty.

Feature block	Δ ROC	95% CI	Δ PR	95% CI
Personal: gender	+0.0526	–	+0.0580	–
Address observability	+0.0465	–	+0.0939	–
Administrative geography	+0.0188	–	+0.0761	–
Physical geography	+0.0000	–	-0.0003	–
Address-record semantics	+0.0090	–	+0.0224	–
Family observability	+0.0217	–	+0.0354	–
Family topology	+0.0023	[0.0019, 0.0026]	+0.0024	[0.0018, 0.0029]
Family capital (full record)	+0.0007	[0.0005, 0.0009]	+0.0011	[0.0007, 0.0015]
Family capital (train observed)	+0.0003	[0.0001, 0.0005]	+0.0005	[0.0002, 0.0009]

(95% CI [-0.0009, 0.0002]) and the train-observed F4–F2 increment was -0.000394 (95% CI [-0.0010, 0.0002]); the corresponding Ming

PR-AUC intervals also included zero. Across populations, family

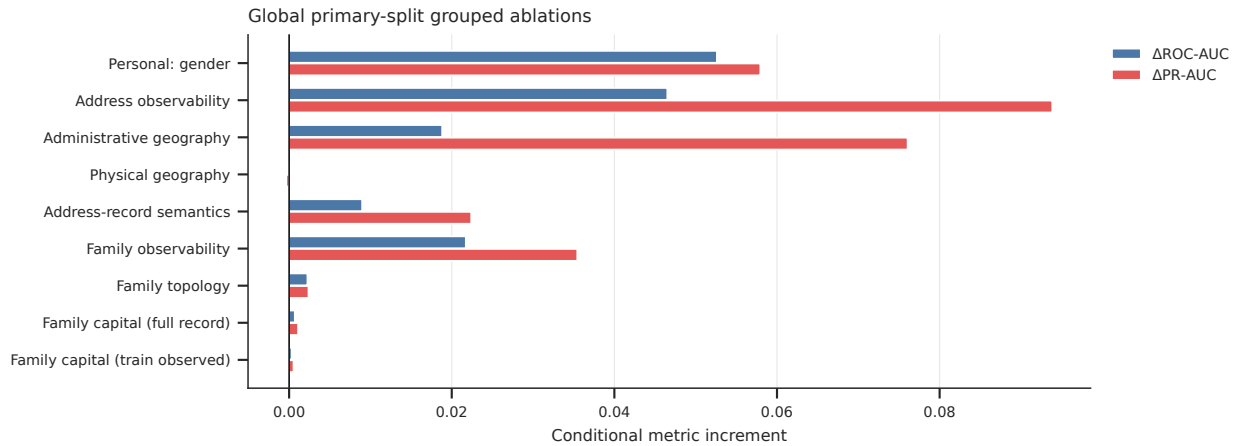


Figure 5: Global grouped ablations. Bars show conditional ROC-AUC and PR-AUC changes for prespecified nested feature blocks. Negative values extend left of zero. Unsupported intervals are omitted because exact paired predictions were not retained for every contrast.

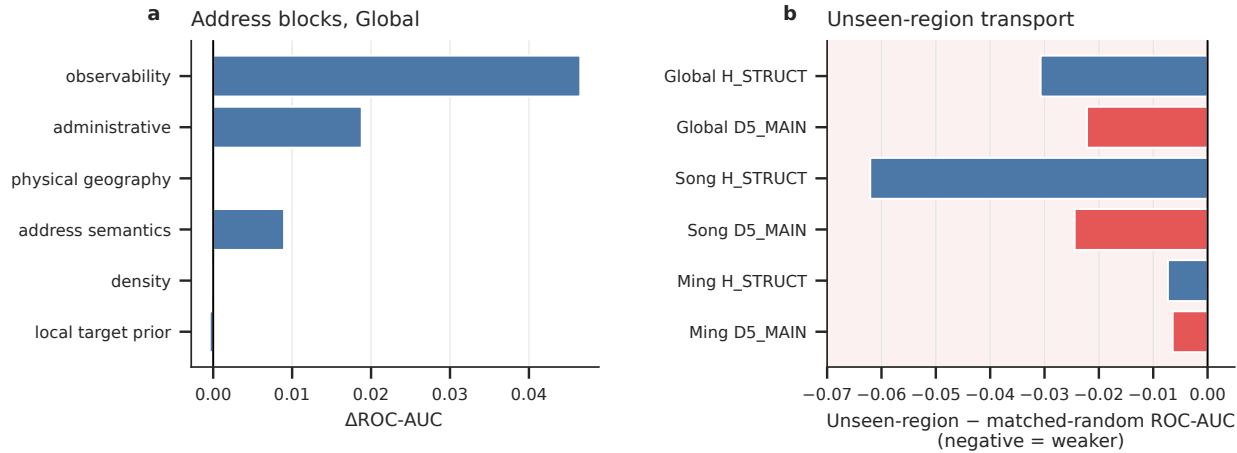


Figure 6: Address decomposition and matched-support spatial transport. a, Global conditional address-block ROC-AUC increments. b, Unseen-region minus matched-random ROC-AUC within reliable-prefecture support. Negative values show weaker transport, not causal geographic effects.

observability and topology contributed much more predictive information than family-capital summaries (Figure 7).

The source audit used marked-primary person-to-source edges without the target. The largest connected component contained 448,632 of 452,035 eligible people (99.25%), exceeding the prespecified 20% ceiling. Full-coverage source-group confirmation was infeasible under the prespecified full-coverage connected-component protocol. This protocol result does not rule out a narrower source-specific validation design. A selective single-primary-source subset could support a narrower source-holdout sensitivity, but it would not represent the full CBDB-covered population.

9 MODEL INTERPRETATION

Grouped SHAP passed additivity checks for all nine population-model pairs (Figure 8). Attribution shifted across populations; re-naming other to historical_regime changed only the display aggregation.

Song's local prior receives substantial SHAP share despite a near-zero conditional increment, and D6_UPPER assigns 2.87% of Global attribution to family capital despite its small increment. Correlated predictors can redistribute attribution, so SHAP is neither independent contribution nor causal effect [11, 14].

10 DISCUSSION

The results establish strong within-database prediction while exposing its documentary basis. H_STRUCT retains observable

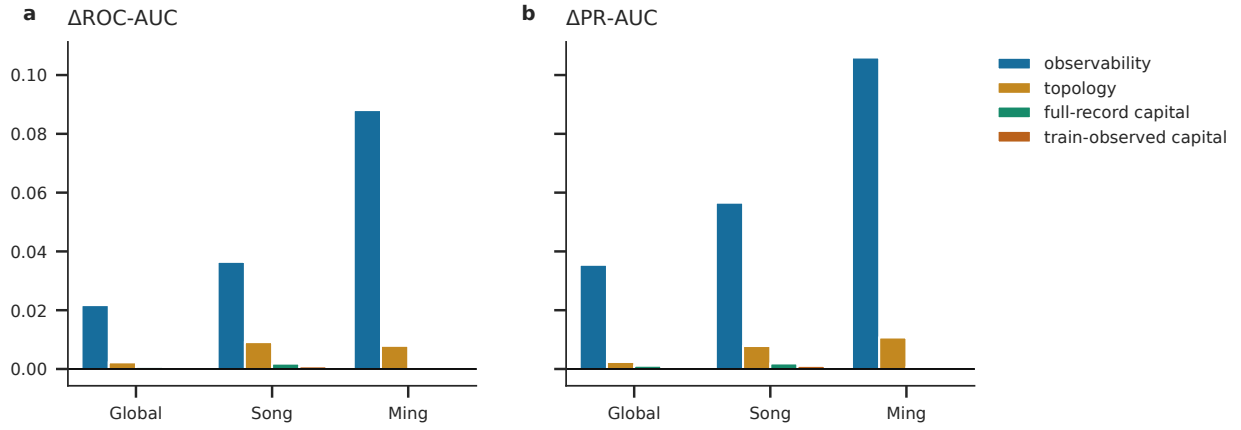


Figure 7: Family-feature decomposition. Conditional ROC-AUC and PR-AUC increments are shown for Global, Song, and Ming. Observability and topology are separated from full-record and train-observed political-capital summaries.

Table 5: Distribution-shift evidence and its availability. Whole-family rows apply only to the frozen F2 structural comparator in Global and Ming; missing performance is labeled explicitly.

Protocol	Population	Model	Shift ROC	ΔROC	Status
Matched-support spatial	Global	D5_MAIN	0.8976	-0.0222	RUN
Matched-support spatial	Global	H_STRUCT	0.8724	-0.0307	RUN
Family-group holdout	Global	F2 structural comparator	0.9272	-0.0022	RUN
Family-group holdout	Ming	F2 structural comparator	0.9090	-0.0032	RUN
SAFE temporal	Global	Locked models	–	–	SPLIT_ONLY_NO_LOCKED_PERFORMANCE
Qing dynasty holdout	Qing	Locked models	–	–	NOT_RUN_NO_FROZEN_HOLDOUT
Source group	Global	H_STRUCT/D5_MAIN	–	–	SOURCE_GROUP_CONFIRMATION_NOT_FEASIBLE

geography and kin structure, D5_MAIN gains from additional documentation-linked signals, and D6_UPPER adds little discrimination from relatives’ lifetime records.

The contribution is an auditable separation of label semantics, structural and documentation-linked increments, family-capital increments, and shift boundaries [9, 12]. Spatial degradation limits random-holdout transport, while whole-family evidence supports only F2 in Global and Ming, not D5_MAIN.

Gender, regime, and family signals may reflect historical stratification, textual survival, editorial selection, encoding, or their combination. Source-specific or temporally anchored designs are needed to distinguish these explanations.

11 LIMITATIONS

Measurement and label validity. E measures heterogeneous ENTRY-record presence, not T ; P is also incomplete. Bootstrap intervals cover test-sample variation for E , not its definition or historical validity.

Selection and documentation bias. CBDB selection varies across period, region, gender, relations, and documentation. Because these processes affect predictors and E , discrimination is neither population prevalence nor historical mechanism. Source confirmation was infeasible under the prespecified full-coverage connected-component protocol.

Temporal and transductive information. D5_MAIN is retrospective, and 9.05% safe-birth coverage prevents broad pre-entry anchoring. Family topology is partly transductive and relatives’ lifetime outcomes are temporally ambiguous, restricting D6_UPPER and family capital to database-internal bounds.

Generalization and adaptive test use. Spatial shift weakened performance, while temporal, Qing, and source-group locked performance is unavailable. Whole-family performance applies only to F2 in Global and Ming, not D5_MAIN. Repeated project-phase test viewing creates adaptation risk; SHAP remains non-causal.

12 CONCLUSION

CBDB ENTRY-record presence is highly predictable within the frozen database. Observability and topology dominate small, Ming-uncertain family-capital increments, and unseen-region transport is weaker. Whole-family evidence is F2-only. These results concern structured records, not true entry or causal office access.

13 REPRODUCIBILITY AND AVAILABILITY

13.1 Data Availability

This study reuses third-party CBDB data from the official release channel [5, 10]. The brief reports about 515,488 people; the verified `cbdb_20260829.sqlite3` release contains 661,124 BIOG_MAIN records. Counts are release-specific, and all results use that release.

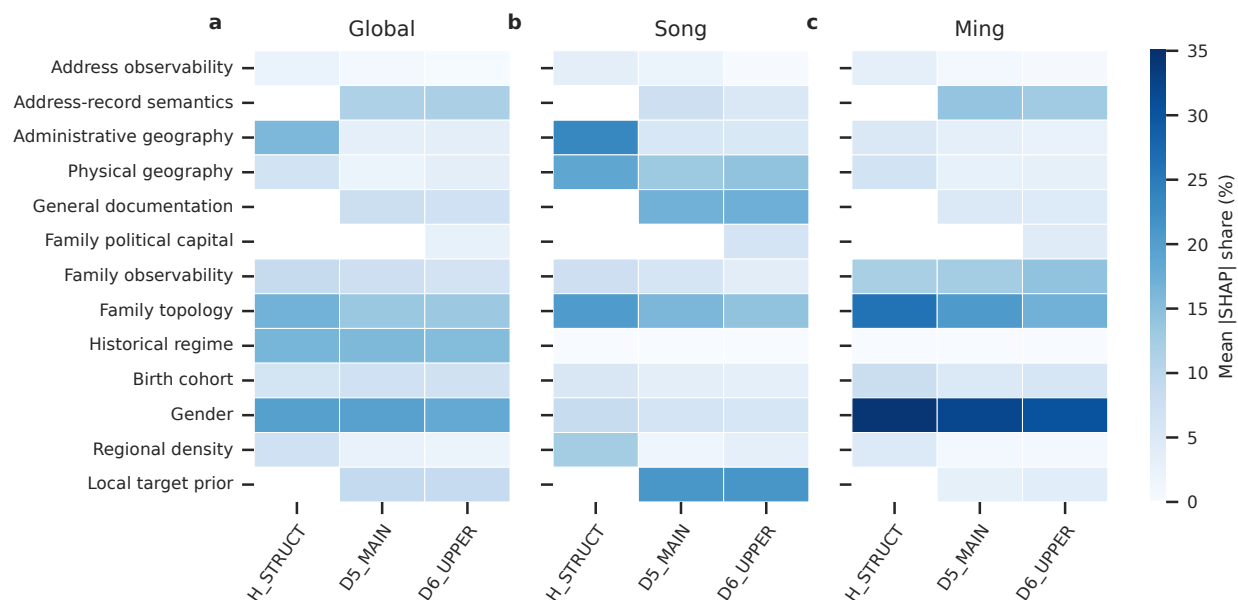


Figure 8: Grouped mean absolute SHAP shares. Global, Song, and Ming panels compare H_STRUCT, D5_MAIN, and D6_UPPER. Physical geography comprises latitude, longitude, and distance to the dynasty-specific capital. Blank cells denote groups absent from a model. Shares are fitted-model attributions, not conditional gains or causal effects.

Its SHA256 is verified. SQLite files are not redistributed; the package provides predictions, tables, figure sources, and regeneration code under CBDB access terms.

13.2 Code Availability

The package contains verification, preprocessing, frozen splits, features, evaluation, figures, environments, and tests. Its MIT License grants no CBDB redistribution rights, and no repository identifier is claimed.

13.3 Reproducibility Statement

Seed 42 and person splits are frozen; test labels selected neither hyperparameters nor calibration. Versions, parameters, hashes, and corrections are recorded. Run bash scripts/run_phase3_1_1_final_polish.sh.

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A TARGET TAXONOMY AND FEATURE BOUNDARIES

The person-level grain begins with BIOG_MAIN. ENTRY and posting relations define E and P , while address, kin, association, status, text, institution, and source relations are aggregated separately before joining. The audited ENTRY taxonomy includes examination or degree, school or student status, recommendation, hereditary or *yin* privilege, military entry, purchase or donation, direct appointment, other clear entry, failed entry, and ambiguous records. A person may occupy several categories.

The global leakage policy forbids identifiers, target fields, ENTRY or posting counts, focal posting presence, raw index-year fields, and unsafe years. Regional density is training-only, and the supervised local prior is out-of-fold on training people. F3 uses full-record relatives' outcomes; F4 restricts outcomes to training-observed relatives but is not strictly pre-entry.

B EXACT LOCAL TARGET-PRIOR CONSTRUCTION

For population-specific training data, let r_i be the exact `addr_id` key for person i ; missing keys are mapped to the literal group `__MISSING__`. With smoothing strength $\alpha = 20$, regional target sum s_r , regional count n_r , and fitting-sample target mean μ , the encoder is

$$\hat{p}_r = \frac{s_r + \alpha\mu}{n_r + \alpha}. \quad (2)$$

Training values use five deterministic out-of-fold partitions. A person's fold is the little-endian integer represented by the first eight bytes of $\text{SHA256}(42|\text{person_id})$, modulo five. For a row held out in fold f , $s_{r,-f}$, $n_{r,-f}$, and μ_{-f} use only other folds; a key unseen outside f receives μ_{-f} . The row's own label and all labels in its fold are therefore excluded.

After producing training OOF values, the encoder is fitted on the complete training partition. Validation and test transformations read region keys but no held-out labels; unseen keys receive the complete-training mean. Encoders are fitted separately for Global, Song, and Ming and refitted within primary, matched-random, and matched-spatial protocols. There is no hierarchical geographic backoff, minimum-count pooling, coordinate-nearest fallback, or external prior. The machine-readable specification is distributed as `local_target_prior_spec.json`.

C ADDITIONAL DIAGNOSTICS

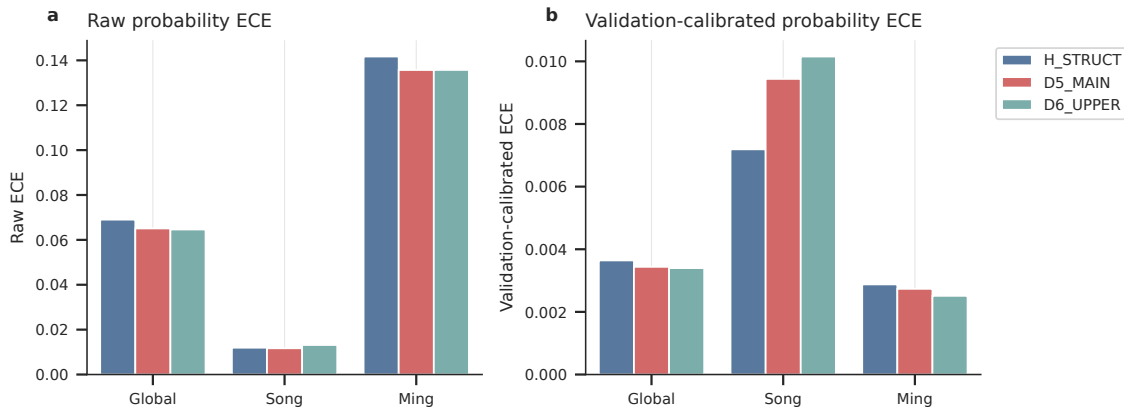


Figure A1: Raw and validation-calibrated ECE. The sigmoid calibrator was fitted on validation predictions only. Lower ECE is better; the panels use different y-axis scales.

D COMPLETE EVALUATION NOTES

Family holdout keeps connected family groups disjoint. Matched spatial comparisons restrict random and spatial tests to identical reliable-prefecture support, although the individuals differ. The SAFE temporal split uses whole years, with training through 1729, validation through 1821, and testing thereafter; 601,273 people are ineligible. Qing results remain descriptive because no frozen Qing holdout exists.

Calibration uses a validation-fitted sigmoid and ten ECE bins. Paired bootstrap intervals use 500 stratified resamples only where exact paired predictions were retained. Grouped SHAP uses 10,000-person samples for each of nine population-model combinations, and all additivity checks pass. The review bundle contains revised grouped, feature-level, and direction summaries with `historical_regime` as the canonical dynasty group.

E FROZEN PARAMETERS AND FULL RESULTS

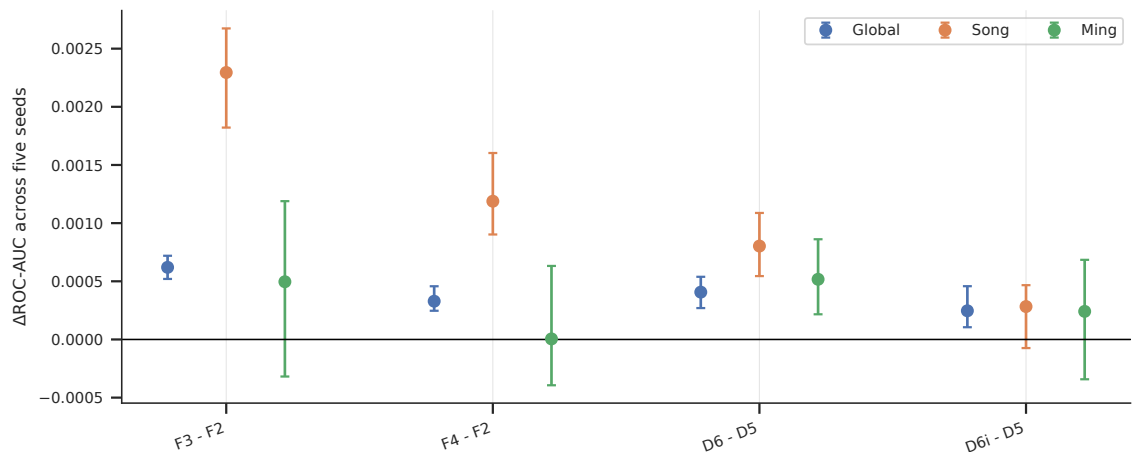


Figure A2: Five-seed stability. Points are mean incremental ROC-AUC values and whiskers span the observed minimum to maximum across five prespecified seeds.

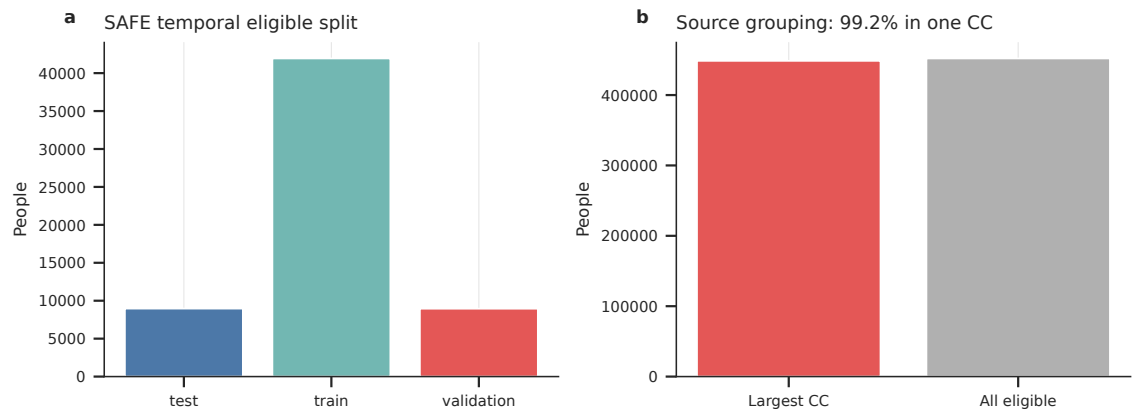


Figure A3: Scope of uncompleted robustness tests. a, Eligible people in the frozen SAFE temporal split, for which no locked performance artifact exists. b, The largest primary-source connected component compared with all eligible people.

Table A1: Frozen CatBoost and evaluation parameters. Validation chooses early stopping, class weighting, and thresholds; the test does not.

Parameter	Frozen value
canonical_seed	42
iterations	500
learning_rate	0.05
depth	7
loss_function	Logloss
eval_metric	AUC
l2_leaf_reg	5
early_stopping_rounds	80
class_weight_candidates	['None', 'Balanced']
bootstrap_resamples	500
calibration_bins	10

Table A1b: Retained operating points for the frozen primary models. Numeric class weights were not serialized; the retained weighting scheme is reported without reconstruction. The sigmoid is $\sigma(a \logit(p)+b)$ and was fitted only on validation predictions.

Population	Model	Selected weighting	Threshold	Objective / selection	Calibrator parameters
Global	Logistic M6	balanced (logistic); numeric weights NOT_RETAINED	0.53	binary logistic loss; lbfgs; frozen linear baseline	NOT_RETAINED
Global	H_STRUCT	Balanced (CatBoost); numeric weights NOT_RETAINED	0.47	CatBoost Logloss; validation ROC-AUC, then lower LogLoss	$a = 1.014113; b = -0.695780$
Global	D5_MAIN	Balanced (CatBoost); numeric weights NOT_RETAINED	0.53	CatBoost Logloss; validation ROC-AUC, then lower LogLoss	$a = 1.023418; b = -0.704666$
Global	D6_UPPER	Balanced (CatBoost); numeric weights NOT_RETAINED	0.49	CatBoost Logloss; validation ROC-AUC, then lower LogLoss	$a = 1.022593; b = -0.704377$
Song	Logistic M6	balanced (logistic); numeric weights NOT_RETAINED	0.53	binary logistic loss; lbfgs; frozen linear baseline	NOT_RETAINED
Song	H_STRUCT	Balanced (CatBoost); numeric weights NOT_RETAINED	0.56	CatBoost Logloss; validation ROC-AUC, then lower LogLoss	$a = 0.996209; b = -0.079308$
Song	D5_MAIN	Balanced (CatBoost); numeric weights NOT_RETAINED	0.52	CatBoost Logloss; validation ROC-AUC, then lower LogLoss	$a = 0.996765; b = -0.103745$
Song	D6_UPPER	Balanced (CatBoost); numeric weights NOT_RETAINED	0.56	CatBoost Logloss; validation ROC-AUC, then lower LogLoss	$a = 1.008454; b = -0.099204$
Ming	Logistic M6	balanced (logistic); numeric weights NOT_RETAINED	0.49	binary logistic loss; lbfgs; frozen linear baseline	NOT_RETAINED
Ming	H_STRUCT	Balanced (CatBoost); numeric weights NOT_RETAINED	0.52	CatBoost Logloss; validation ROC-AUC, then lower LogLoss	$a = 1.028157; b = -1.377243$
Ming	D5_MAIN	Balanced (CatBoost); numeric weights NOT_RETAINED	0.50	CatBoost Logloss; validation ROC-AUC, then lower LogLoss	$a = 1.049390; b = -1.359069$
Ming	D6_UPPER	Balanced (CatBoost); numeric weights NOT_RETAINED	0.47	CatBoost Logloss; validation ROC-AUC, then lower LogLoss	$a = 1.052307; b = -1.359874$

Table A2: Full-precision locked primary-test metrics. Prevalence accompanies PR-AUC; ECE follows validation-fitted sigmoid calibration.

Population	Model	Prev.	ROC	PR	Bal. Acc.	F1	LogLoss	Brier	ECE
Global	Logistic M6	0.3339	0.916038	0.843596	0.848544	0.792425	0.363532	0.112821	–
Global	H_STRUCT	0.3339	0.926721	0.858211	0.860891	0.809549	0.339298	0.104816	0.003644
Global	D5_MAIN	0.3339	0.936956	0.882535	0.869045	0.820769	0.315106	0.097356	0.003438
Global	D6_UPPER	0.3339	0.937421	0.883128	0.869739	0.821732	0.313950	0.096938	0.003398
Song	Logistic M6	0.4827	0.900769	0.880995	0.833327	0.831659	0.398458	0.122321	–
Song	H_STRUCT	0.4827	0.918350	0.911650	0.847816	0.842489	0.360396	0.111199	0.007191
Song	D5_MAIN	0.4827	0.938351	0.939838	0.870885	0.865457	0.308890	0.093986	0.009439
Song	D6_UPPER	0.4827	0.939798	0.941195	0.871448	0.866054	0.305452	0.092799	0.010154
Ming	Logistic M6	0.1993	0.902058	0.776510	0.801988	0.696932	0.349956	0.112626	–
Ming	H_STRUCT	0.1993	0.910657	0.779909	0.813006	0.712249	0.346308	0.109660	0.002878

Population	Model	Prev.	ROC	PR	Bal. Acc.	F1	LogLoss	Brier	ECE
Ming	D5_MAIN	0.1993	0.916395	0.800527	0.816932	0.726957	0.322947	0.102572	0.002736
Ming	D6_UPPER	0.1993	0.916696	0.801196	0.817768	0.727861	0.322401	0.102311	0.002513

Table A3: All grouped ablation contrasts. Intervals are shown only where the exact paired predictions were retained.

Population	Block	Contrast	Δ ROC	ROC CI	Δ PR	PR CI
Global	Personal: gender	P1 - P0	+0.052609	–	+0.057951	–
Song	Personal: gender	P1 - P0	+0.047146	–	+0.024770	–
Ming	Personal: gender	P1 - P0	+0.112030	–	+0.043581	–
Global	Address observability	A1 - A0	+0.046510	–	+0.093855	–
Song	Address observability	A1 - A0	+0.200425	–	+0.147162	–
Ming	Address observability	A1 - A0	+0.054365	–	+0.047488	–
Global	Administrative geography	A2 - A1	+0.018820	–	+0.076062	–
Song	Administrative geography	A2 - A1	+0.102003	–	+0.166421	–
Ming	Administrative geography	A2 - A1	+0.016509	–	+0.090486	–
Global	Physical geography	A3 - A2	+0.000014	–	-0.000343	–
Song	Physical geography	A3 - A2	+0.000311	–	+0.000883	–
Ming	Physical geography	A3 - A2	+0.000152	–	+0.002470	–
Global	Address-record semantics	A4 - A3	+0.008980	–	+0.022396	–
Song	Address-record semantics	A4 - A3	+0.000514	–	+0.001088	–
Ming	Address-record semantics	A4 - A3	+0.030118	–	+0.099341	–
Global	Family observability	F1 - F0	+0.021728	–	+0.035438	–
Song	Family observability	F1 - F0	+0.036438	–	+0.056570	–
Ming	Family observability	F1 - F0	+0.088076	–	+0.105958	–
Global	Family topology	F2 - F1	+0.002251	[0.0019,0.0026]	+0.002381	[0.0018,0.0029]
Song	Family topology	F2 - F1	+0.009136	[0.0074,0.0106]	+0.007811	[0.0060,0.0095]
Ming	Family topology	F2 - F1	+0.007871	[0.0064,0.0094]	+0.010709	[0.0087,0.0128]
Global	Family capital (full record)	F3 - F2	+0.000701	[0.0005,0.0009]	+0.001098	[0.0007,0.0015]
Song	Family capital (full record)	F3 - F2	+0.001821	[0.0009,0.0027]	+0.001877	[0.0006,0.0031]
Ming	Family capital (full record)	F3 - F2	-0.000319	[-0.0009,0.0002]	+0.000576	[-0.0004,0.0015]
Global	Family capital (train observed)	F4 - F2	+0.000311	[0.0001,0.0005]	+0.000541	[0.0002,0.0009]
Song	Family capital (train observed)	F4 - F2	+0.000902	[0.0002,0.0016]	+0.001049	[0.0002,0.0019]
Ming	Family capital (train observed)	F4 - F2	-0.000394	[-0.0010,0.0002]	+0.000525	[-0.0004,0.0014]

Table A4: Complete distribution-shift evidence. Dashes preserve experiments that were unavailable or infeasible rather than inventing estimates.

Population	Model	Protocol	Shift ROC	Δ ROC	Shift PR	Δ PR	Status
Global	D5_MAIN	Matched-support spatial	0.897571	-0.022177	0.879052	-0.035853	RUN
Global	H_STRUCT	Matched-support spatial	0.872439	-0.030714	0.831291	-0.060427	RUN
Song	D5_MAIN	Matched-support spatial	0.893811	-0.024455	0.933361	-0.026770	RUN
Song	H_STRUCT	Matched-support spatial	0.810654	-0.062065	0.852220	-0.077701	RUN
Ming	D5_MAIN	Matched-support spatial	0.932692	-0.006423	0.920740	-0.005574	RUN
Ming	H_STRUCT	Matched-support spatial	0.922810	-0.007297	0.895060	-0.010600	RUN
Global	F2 structural comparator	Family-group holdout	0.927159	-0.002154	0.866816	-0.001824	RUN

Population	Model	Protocol	Shift ROC	Δ ROC	Shift PR	Δ PR	Status
Ming	F2 structural comparator	Family-group holdout	0.908970	-0.003194	0.789235	-0.003727	RUN
Global	Locked models	SAFE temporal	–	–	–	–	SPLIT_ONLY_NO_LOCKED_PERFORMANCE
Qing	Locked models	Qing dynasty holdout	–	–	–	–	NOT_RUN_NO_FROZEN_HOLDOUT
Global	H_STRUCT/D5_MAIN	Source group	–	–	–	–	SOURCE_GROUP_CONFIRMATION_NOT_FEASIBLE

F COMPLETE FROZEN FEATURE REGISTRY

Table A5: Complete frozen feature registry. Repeated features document the exact membership of each locked model.

Model	Feature	Role
H_STRUCT	dynasty_name	categorical
H_STRUCT	gender	categorical
H_STRUCT	safe_birth_decade	categorical
H_STRUCT	province_id	categorical
H_STRUCT	prefecture_id	categorical
H_STRUCT	county_id	categorical
H_STRUCT	has_safe_birth_year	numeric
H_STRUCT	has_geography	numeric
H_STRUCT	has_valid_coordinates	numeric
H_STRUCT	latitude	numeric
H_STRUCT	longitude	numeric
H_STRUCT	distance_to_dynasty_capital_km	numeric
H_STRUCT	train_region_person_count	numeric
H_STRUCT	train_region_log_density	numeric
H_STRUCT	has_kin	numeric
H_STRUCT	has_core_family	numeric
H_STRUCT	father_identified	numeric
H_STRUCT	mother_identified	numeric
H_STRUCT	any_grandfather_identified	numeric
H_STRUCT	n_known_parents	numeric
H_STRUCT	n_known_grandparents	numeric
H_STRUCT	n_known_ancestors	numeric
H_STRUCT	n_known_same_generation	numeric
H_STRUCT	n_known_descendants	numeric
H_STRUCT	n_known_core_kin	numeric
H_STRUCT	family_group_size	numeric
D5_MAIN	dynasty_name	categorical
D5_MAIN	gender	categorical
D5_MAIN	safe_birth_decade	categorical
D5_MAIN	province_id	categorical
D5_MAIN	prefecture_id	categorical
D5_MAIN	county_id	categorical
D5_MAIN	addr_type_name	categorical
D5_MAIN	has_address	numeric
D5_MAIN	has_kin	numeric
D5_MAIN	has_assoc	numeric
D5_MAIN	has_status	numeric
D5_MAIN	has_text	numeric
D5_MAIN	has_institution	numeric

Model	Feature	Role
D5_MAIN	documentation_intensity	numeric
D5_MAIN	has_safe_birth_year	numeric
D5_MAIN	has_geography	numeric
D5_MAIN	has_valid_coordinates	numeric
D5_MAIN	latitude	numeric
D5_MAIN	longitude	numeric
D5_MAIN	distance_to_dynasty_capital_km	numeric
D5_MAIN	train_region_person_count	numeric
D5_MAIN	train_region_log_density	numeric
D5_MAIN	local_target_prior	numeric
D5_MAIN	has_core_family	numeric
D5_MAIN	father_identified	numeric
D5_MAIN	mother_identified	numeric
D5_MAIN	any_grandfather_identified	numeric
D5_MAIN	n_known_parents	numeric
D5_MAIN	n_known_grandparents	numeric
D5_MAIN	n_known_ancestors	numeric
D5_MAIN	n_known_same_generation	numeric
D5_MAIN	n_known_descendants	numeric
D5_MAIN	n_known_core_kin	numeric
D5_MAIN	family_group_size	numeric
D6_UPPER	dynasty_name	categorical
D6_UPPER	gender	categorical
D6_UPPER	safe_birth_decade	categorical
D6_UPPER	province_id	categorical
D6_UPPER	prefecture_id	categorical
D6_UPPER	county_id	categorical
D6_UPPER	addr_type_name	categorical
D6_UPPER	has_address	numeric
D6_UPPER	has_kin	numeric
D6_UPPER	has_assoc	numeric
D6_UPPER	has_status	numeric
D6_UPPER	has_text	numeric
D6_UPPER	has_institution	numeric
D6_UPPER	documentation_intensity	numeric
D6_UPPER	has_safe_birth_year	numeric
D6_UPPER	has_geography	numeric
D6_UPPER	has_valid_coordinates	numeric
D6_UPPER	latitude	numeric
D6_UPPER	longitude	numeric
D6_UPPER	distance_to_dynasty_capital_km	numeric
D6_UPPER	train_region_person_count	numeric
D6_UPPER	train_region_log_density	numeric
D6_UPPER	local_target_prior	numeric
D6_UPPER	has_core_family	numeric
D6_UPPER	father_identified	numeric
D6_UPPER	mother_identified	numeric
D6_UPPER	any_grandfather_identified	numeric
D6_UPPER	n_known_parents	numeric
D6_UPPER	n_known_grandparents	numeric
D6_UPPER	n_known_ancestors	numeric
D6_UPPER	n_known_same_generation	numeric
D6_UPPER	n_known_descendants	numeric
D6_UPPER	n_known_core_kin	numeric
D6_UPPER	family_group_size	numeric

Model	Feature	Role
D6_UPPER	father_ever_entry	numeric
D6_UPPER	father_ever_posting	numeric
D6_UPPER	paternal_grandfather_ever_entry	numeric
D6_UPPER	paternal_grandfather_ever_posting	numeric
D6_UPPER	maternal_grandfather_ever_entry	numeric
D6_UPPER	maternal_grandfather_ever_posting	numeric
D6_UPPER	n_older_kin_ever_entry	numeric
D6_UPPER	n_older_kin_ever_posting	numeric
D6_UPPER	older_kin_entry_ratio	numeric
D6_UPPER	older_kin_posting_ratio	numeric

G REPRODUCTION COMMAND

From the project root, run `bash scripts/run_phase3_1_1_final_polish.sh`. The runner verifies the Phase 3.1 baseline, rebuilds only Phase 3.1.1 tables and figures, compiles both manuscripts, renders every PDF page, runs audits and tests, packages the deliverables, and validates both archives.